is unaffected by its overall phase-factor) eliminates a second.

- Having chosen $|u\rangle$ and $|d\rangle$ as our main basis vectors, we figured out how to represent the other two pairs of basis vectors as linear combinations of $|u\rangle$ and $|d\rangle$, using additional orthogonality and probability-based constraints.

- Finally, we established a way to represent our main basis vectors as columns. This representation is not unique. In the next lecture, we'll use our $|u\rangle$ and $|d\rangle$ column vectors to derive column vectors for the two other bases.

While achieving these concrete results, we got a chance to see some state-vector mathematics in action and learn something about how these mathematical objects correspond to physical spins. Although we will focus on spin, the same concepts and techniques apply to other quantum systems as well. Please take a little time to assimilate the material we've covered so far before moving on to the next lecture. As I said at the beginning, it will really pay off.